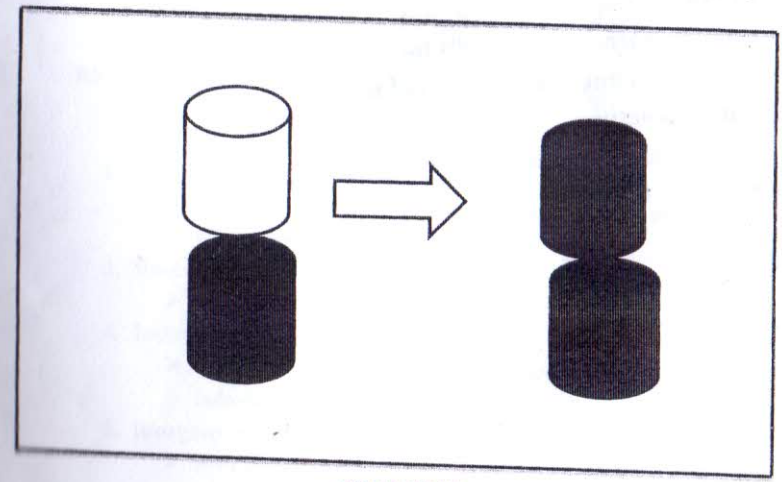
# JUNIOR SECONDARY

TAONGA kl. KALINGA

# CHEMISTRY



**TORP**

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# CHEMISTRY

## BOOK ONE

Topics

- Introduction to chemistry
- Essential mathematical skills for chemistry
- Composition and classification of matter
- Atomic structure
- Physical and chemical changes
- Organic compounds

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## CHAPTER 1

## INTRODUCTION TO CHEMISTRY

- ***Chemistry*** is a branch of science that deals with the identification of the substance of which matter is composed.
- ***Chemists*** are people who study chemistry. They do much of their work in laboratories using chemicals and apparatus.
- Some of the apparatus includes beakers, test tubes, wash bottles, measuring cylinder, tongs, thermometers, stop watches, conical flasks among others.

### CHEMISTRY AND ITS BRANCHES

- ***Chemistry*** is defined as the study of the composition, properties and uses of all the material that make up our world both living and non-living.

### BRANCHES OF CHEMISTRY

1. **Physical Chemistry**
   - Physical principles to understand properties of matter.
2. **Analytical Chemistry**
   - Chemistry of matter and development of tools used to measure properties of matter.
3. **Environmental Chemistry**
   - Studies the environment.
4. **Industrial Chemistry**
   - Chemistry which brings the link between the research and industrial – scale engineering.
5. **Inorganic Chemistry**
   - Studies compounds which do not contain carbon – hydrogen bonds.
6. **Organic Chemistry**
   - Deals with compounds that contain carbon – hydrogen bonds.

☞ **Chemistry** is important because it is used in everyday life.

## CHEMISTRY AND CAREERS

- There are several careers in chemistry such as
  1. **Biochemists**
     - Chemistry dealing with structure, composition and chemical reactions of living things.
  2. **Teacher chemist**
     - Chemists in educational institutions.
  3. **Geochemists**
     - Chemists dealing with rocks.
  4. **Pharmacists**
     - Chemistry in pharmacies.
  5. **Environmental chemists**
     - Chemists who study the environment.
  6. **Quality control and Assurance personnel**
     - Chemistry in industries and companies such as Malawi Bureau of standards.
  7. **Food chemists**
     - Involved in analyzing food items.
  8. **Photo chemists**
     - Specialize in photochemistry (the study of chemical reactions involving sunlight).

## SAFETY IN THE LABORATORY

The laboratory is a place or room where chemicals, apparatus and other materials are kept. Such things can cause harm such as:

- Electricity can hurt and kill.
- Pieces of broken glass can pierce through the skin.



- Heat and hot objects can cause burns and scalds.
- Heavy objects may cause bruises.
- Loose clothing can cause apparatus to fall or catch fire.
- Long hair can catch fire.

## LABORATORY RULES

1. Do not drink, taste or eat anything in the laboratory because chemicals are toxic.
2. Wear protective clothing when handling chemicals.
3. Tie long hair.
4. Carry apparatus and do not throw.
5. If acid splashes over your skin, wash in plenty of cold water.
6. Follow the procedures of the experiment strictly.
7. After an experiment wash your hands and clean the bench and apparatus which was used for the experiment.

## PROPER USE OF PROTECTIVE EQUIPMENT IN A CHEMISTRY LABORATORY

1. GOGGLES: Protect eyes from fumes.
2. LAB COATS: Protect an individual from corrosive and harmful substances such as acidic solutions.
3. GLOVES: Protect hands when working with solutions.
4. GAS MASKS: They absorb poisonous gases.
5. FIRE EXTINGUISHER: Used to put off fire during a fire accident.

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## HAZARD SYMBOLS

| Symbol | Meaning | Interpretation |
|---|---|---|
|  | Harmful / Irritant substance | • Avoid skin an eye contact. |
|  | Risk of ionizing radiation | • Wear protective clothing to avoid skin contact.<br>• Store in special containers or places. |
|  | Corrosive substance | • Avoid skin contact and with clothes by wearing gloves, dustcoats and aprons. |

## OTHER HAZARDS

1. Heating with a Bunsen burner. If gas is not controlled the whole lab can catch fire.
2. Handling glassware apparatus. Sometimes the glassware can have cracks which can pierce through or cut the skin.

## SCIENTIFIC METHOD OF INVESTIGATION

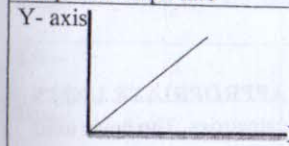
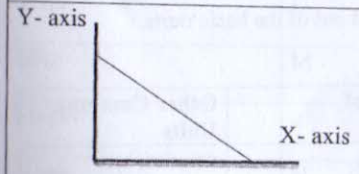
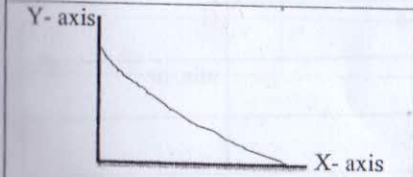
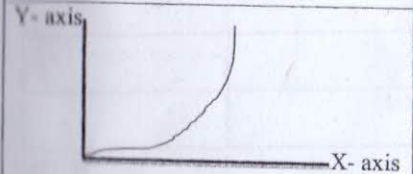
- A scientific inquiry is important because it helps to RESEARCH, INVESTIGATE and EXPLORE facts in order to DEVELOP CONCEPTS, KNOWLEDGE and SKILLS.
- Any scientific inquiry follows a properly designed procedure called a **scientific method** in order to come up with accurate results.



## STAGES IN SCIENTIFIC INQUIRY

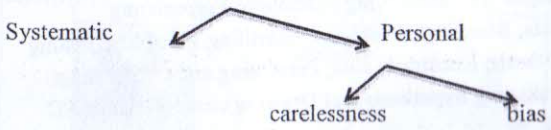
- These stages are: Identifying a problem; Formulating hypothesis; Selecting variables; Controlling variables; Testing the hypothesis; Recording data; Presenting and interpreting data; Evaluating hypothesis and Drawing conclusions [IFSC TRPED].

## SHAPES OF GRAPHS AND THEIR MEANINGS

| Shape of Graph line | Meaning |
|---|---|
|  | As the independent variable increases or decreases, the dependent variable does the same in the same proportion. |
|  | As the independent variable increases, the dependent variable decreases in the same proportion. |
|  | As the independent variable increases, the dependent variable decreases exponentially. |
|  | As the independent variable increases, the dependent variable increases exponentially. |

## TYPES OF EXPERIMENTAL ERRORS

Errors




1) Systematic errors: Occur through faulty apparatus.
2) Personal (a)carelessness: Arise through carelessness in reading and recording data.

(b) bias: These arise through bias i.e. what you want the results to be like.

## PRESENTING RESULTS USING APPROPRIATE UNITS

- The units are BASIC and DERIVED categories. The basic units represent each of the 7 basic quantities that exist in nature. Derived units are units worked out of the basic units.

**Table**

| Basic Quantity | Unit of Measurement | Symbol | Other Common Units |
|---|---|---|---|
| Length | Metre | M | cm, dm, km |
| Mass | Kilogram | Kg | G |
| Time | Second | S | min, hr |
| Temperature | Kelvin | K | |
| Light intensity | Candela | Cd | |
| Electric current | Ampere | A | |
| Amount of a Substance | Mole | Mol | |



## METRIC UNITS

- These are related by power of 10 as outlined in the table below.

| Prefix | Symbol | Meaning |
|---|---|---|
| Tera - | T | $10^{12}$ |
| Giga- | G | $10^{9}$ |
| Mega- | M | $10^{6}$ |
| Kilo- | K | $10^{3}$ |
| Deci- | D | $10^{-1}$ |
| Centi- | C | $10^{-2}$ |
| Milli- | M | $10^{-3}$ |
| Micro- | μ | $10^{-6}$ |
| Nano- | N | $10^{-9}$ |
| Pico- | P | $10^{-12}$ |

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## DERIVED UNITS

**Table**

| Quantity | Symbol | Quantities Derived from | Units | Other units |
|---|---|---|---|---|
| Area | A | Length × Width (L×W) | $m^2$ | – |
| Volume | V | L × W × H | $m^3$ | – |
| Speed | V | $\frac{distance}{time}$ | m/s | – |
| Acceleration | A | $\frac{speed}{time}$ | $m/s^2$ | – |
| Force | F | mass × acceleration | $Kgm/s^2$ | Newton (N) |
| Pressure | P | $\frac{force}{area}$ | $N/m^2$ | Pascal (pa) |
| Energy (force) | X | Distance | Nm | Joule (J) |
| Density | | $\frac{mass}{volume}$ | $Kg/m^3$ | – |

# CHAPTER 2

## ESSENTIAL MATHEMATICAL SKILLS FOR CHEMISTRY

- Analytical chemistry involves separating, identifying and determining the relative amounts of the components of a substance or part of it called a SAMPLE or ANALYTE. The amounts are measured, either by a measuring cylinder, pipette, syringe, burette, or graduated [calibrated] beaker. These quantities are represented numerically.



## NATURAL FORM, SCIENTIFIC NOTATION AND SIGNIFICANT FIGURES

1. **Natural Form**
   - The natural form of a number is simply how it is represented, e.g. 2, 3,10,140 e.t.c.
2. **Scientific Notation**
   - This is also called STANDARD FORM.
   - It is a brief method of writing very large numbers.
   - The number is expressed into two factors.

     a) The first factor is a number between 1 and 9, called a ***Significant Digit***.

     b) The second is expressed as a Power of 10

     e.g. $3 = 3.0 \times 10^0$; $247 = 2.47 \times 10^2$; $0.000149 = 1.49 \times 10^{-4}$

   - When the decimal point does not move, then the power of 10 is ***ZERO***.
   - When the decimal point moves to the left, the power of 10 is ***positive***.
   - When the decimal point moves to the right, the power of 10 is ***negative***.
   - Zeros between digits are significant.

3. **Significant Figures**

   Steps in writing significant figures:
   - The first non- zero digit is significant.
   - Zeros between non- zero digits are also significant.
   - Zeros after non- zero digits are significant if they are meant to keep place value.
   - Zeros after the decimal point can be left out.

*Worked Example*

a) Write 100945 to 3 significant figures.

***Solution***

100945 = 101000

b) Write 0.0645 to 1 significant figure.

***Solution***

0.0645 = 0.06

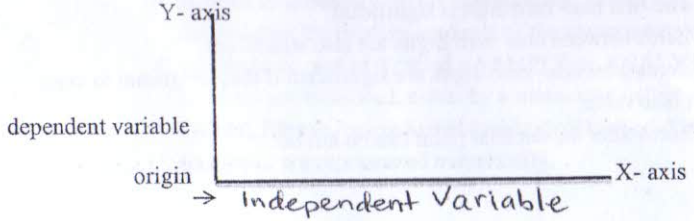
## ACCURACY AND PRECISION

- ***Accuracy*** means the degree to which a measured quantity is very close to the true value. For example, if a true value is 6 then a number very close to 6 can be an accurate number i.e. 5.9 or 6.1.
- ***Precision*** shows how close to one another are replicate [repeated] measurements taken in the same manner. For example, if the results are 5.8, 6.0, 5.9, 6.1, then they indicate precision [the results are almost exact].

## GRAPHICAL PRESENTATION OF DATA

➢ Numerical data can be represented graphically.
There are 3 common types of graphs:

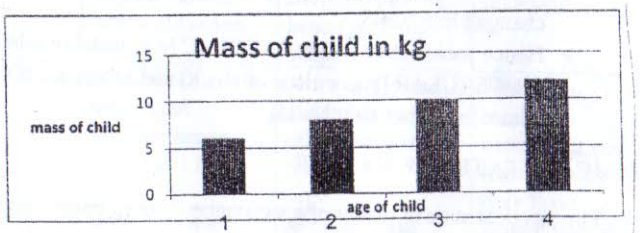
1. LINE GRAPHS
   - These have two axes: Y-axis which represents dependent variable and X-axis which represents the independent variable.



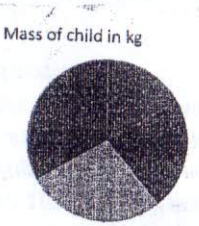



~~independent variable~~

2. BAR GRAPHS
   ➢ These graphs have RECTANGULAR vertical bars representing the dependent variable plotted on the vertical axis against the independent variable which is plotted on the horizontal axis.




3. PIE CHARTS [GRAPHS]
   ➢ These are circular graphs. The circle is divided up proportional to the fraction each data frequency makes on the whole. Each data frequency is taken as a fraction of $360^0$ [angles at a point].




# CHAPTER 3

## COMPOSITION AND CLASSIFICATION OF MATTER

- Matter is anything that has mass and occupies space [or has volume].

- Mass is the quantity of matter in an object.

## COMPOSITION OF MATTER

- Matter is made up of very minute particles called Atoms.
- Atoms can be broken down into smaller particles.
- Atoms chemically combine to form molecules and some atoms can be changed into IONS.
- Hence some substances are ATOMIC [e.g. metals]; others are MOLECULAR [e.g. a piece of chalk] and others are IONIC e.g. common kitchen salt [Na cl].

## PHYSICAL STATES OF MATTER

- The physical state of matter of a substance at room temperature depends on the melting and boiling points.

**There are 3 states of matter**

- Solids
- Liquids
- Gases

## LOOKING AT CHANGES OF FORMS OF MATTER

| CHARACTERISTICS | SOLIDS | LIQUIDS | GASES |
|---|---|---|---|
| Distance between particles. | Very close to each other with little spaces | Fairly close to each other with larger spaces | Far apart from each other hence compressible and |

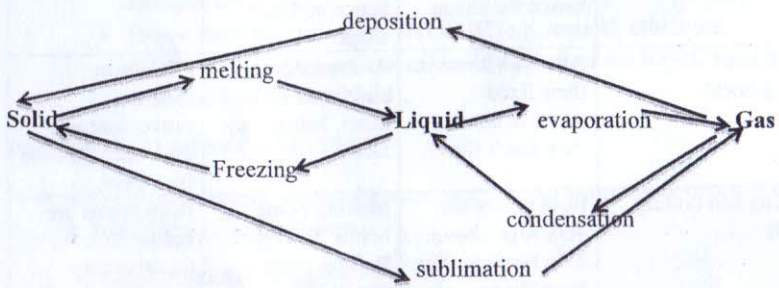


| | | | |
|---|---|---|---|
| | between them hence incompressible, fixed volume and definite shape. | between them compared to solids hence incompressible and fixed volume. | no fixed shape or volume [take shape and volume of a container]. |
| Pattern of arrangement | Arranged in regular pattern hence the shape. | Arranged in irregular pattern hence no fixed shape. | No pattern hence no shape. |
| Mobility of arrangement | Always vibrate in their fixed position hence they don't flow. | Always slip and slide over each other, hence they flow. | Always in random movement and high speed, hence they flow. |
| Melting and boiling points | Both points are very high above $25^0$c because they have strong imfs hence their state. | Melting point below $25^0$c Boiling point above $25^0$c because they have fairly strong imfs hence their state. | Both points are below $25^0$c because they have very weak intermolecular forces (imfs) hence their state. |

- $25^0$c is the room temperature.
- When a solid is heated, the particles gain heat energy and their kinetic energy increase and they start vibrating vigorously. The vibrations become so violent such that the intermolecular forces [imfs] weaken and particles start moving away from each other. The solid then melts and becomes a liquid.
- The same happens with particles of liquid when they are heated. The intermolecular forces [imfs] weaken and the liquid becomes a gas. This is called EVAPORATION or BOILING.
- The temperature at which melting occurs is called MELTING POINT and the temperature at which boiling occurs is called BOILING POINT.
- Cooling a substance removes heat energy and reduces the kinetic energy of the particles. The particles then start vibrating slowly. As such a gas changes to liquid-CONDENSATION and a liquid changes to a solid-FREEZING.

- The temperature at which condensation occurs is called CONDENSATION POINT and at which freezing occurs is called FREEZING POINT.

**Summary of changes of states of matter**



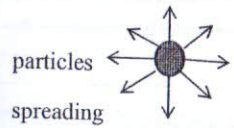

- Melting is also called LIQUEFACTION and freezing is also called SOLIDIFICATION.
- Substances which can **sublime** are iodine and solid carbon dioxide [$CO_2$].

**Existence of molecules in a substance**

- Evidence of the existence of molecules in a substance is seen in the process of DIFFUSION.
- **Diffusion** is the spreading or net movement of particles from a region where they are more concentrated to a region where they are less concentrated.



- Diffusion carries small particles from one region to another.




☞ Practically you experience diffusion when you smell chemicals, perfumes or food. The smell travels by diffusion.

## ELEMENTS AND COMPOUNDS

- Elements are substances that are made up of the same kind of the atoms. e.g. $O_2$, $H_2$, $Br_2$ etc.
- Compounds are substances that are made up of more than one kind of an atom e.g. $H_2O$.
- There are more than 100 different types of atoms and the first 20 of them are:

| Elements | Symbol | Elements | Symbol |
|---|---|---|---|
| Hydrogen | H | Sodium | Na |
| Helium | He | Magnesium | Mg |
| Lithium | Li | Aluminium | Al |
| Oxygen | O | Silicon | Si |
| Beryllium | Be | Phosphorous | P |
| Fluorine | F | Sulphur | S |
| Boron | B | Chlorine | Cl |
| Carbon | C | Argon | Ar |
| Neon | Ne | Potassium | K |

| | | | |
|---|---|---|---|
| Nitrogen | N | Calcium | Ca |

**OTHER COMMON ELEMENTS ARE**

| ELEMENT | SYMBOL |
|---|---|
| Copper | Cu |
| Iron | Fe |
| Lead | Pb |
| Silver | Ag |
| Gold | Au |
| Mercury | Hg |
| Zinc | Zn |

- Substances that are composed of single atoms are MONO-ATOMIC e.g. metals.
- Elements with two same atoms are called DIATOMIC e.g. hydrogen gas, [$H_2$], oxygen gas [$O_2$] etc.
- A majority of substances are compounds.

**TABLE**

| **Diatomic Substances** | **Compounds** |
|---|---|
| Hydrogen ($H_2$) | Carbon dioxide ($CO_2$) |
| Oxygen ($O_2$) | Nitrogen dioxide ($NO_2$) |
| Chlorine ($Cl_2$) | Water ($H_2O$) |
| Nitrogen ($N_2$) | Methane ($CH_4$) |
| Fluorine ($F_2$) | Ammonia ($NH_3$) |
| Ozone ($O_3$) | Carbon monoxide (CO) |
| Bromine ($Br_2$) | Ethanol ($C_2H_5OH$) |
| | Sugar ($C_{12}H_{22}O_{11}$) |

**CHEMICAL OR MOLECULAR FORMULA**

➢ These will show the type and number of atoms in a molecule. For example, $H_2O$ has 2 types of atoms and a total of 3 atoms; $C_{11}$ $H_{22}$ $O_{11}$ has 3 types of atoms and a total of 44 atoms.



**PURE SUBSTANCES AND MIXTURES**

- A pure substance consists of one substance only.
- A mixture is a combination of different types of substances.

**TYPES OF MIXTURES**

➢ There are two different types:

A] HOMOGENOUS MIXTURE

- This is the mixture which has two or more substances uniformly distributed throughout its volume and has the same uniform appearance and composition throughout. Most of these are SOLUTIONS or CLEAR MIXTURES e.g. Fanta.

B] HETEROGENOUS MIXTURE

- This is the mixture which has two or more substances not evenly distributed. Examples are SUSPENSIONS e.g. a suspension of maize flour in water.

**SOLUTION**

- This is a homogenous mixture of one or more solutes dissolved in a solvent.
- A **SOLUTE** is the substance that dissolves to produce a homogenous mixture. e.g. Brine [salt solution].
- A **SOLVENT** is the liquid in which a solute dissolves to produce a homogenous mixture e.g. water.
- Substances which do not dissolve in a particular solvent are called **INSOLUBLE** in that liquid.
- When a substance dissolves in water, the solution is called an **AQUEOUS SOLUTION.**

| SOLUTE | SOLVENT (S) |
|---|---|
| Salt, sugar, $CuSO_4$, $C_2H_5OH$ | Water |
| Paint | Petrol/paraffin |
| Gross paint | White spirit |
| Naphthalene | Turpentine |
| Grease | Propanone/tetrachloro-methane |
| Glues, printing inks | Ethanol |
| | |

**TYPES OF SOLUTIONS**

1. A SOLID IN A LIQUID
   - A solute completely dissolves in a solvent. When the solvent is water then the solution is called AQUEOUS SOLUTION.
2. A LIQUID IN A LIQUID
   - When the liquids dissolve into each other, the liquids are called MISCIBLE LIQUIDS e.g. ethanol in water.
3. A SOLID IN A LIQUID
   - A solution of a metal in a metal is called an ALLOY e.g. steel is an alloy of carbon and iron.
4. A GAS IN A LIQUID
   - For example $CO_2$ and $O_2$ dissolve in water. An example is coca-cola.
5. A GAS IN A GAS
   - For air is a mixture of gases. Different gases dissolve in each other to form air.

**FACTORS THAT AFFECT SOLUBILITY**

- **Temperature**
  - The higher the temperature, the higher the solubility.
- **Size of particles**



  - The smaller the particles, the higher the solubility.
- **Volume of solvent**
  - The more the volume, the higher the solubility.

**SEPARATING MIXTURES**

- Several methods are used.

A] SEPARATING SOLID-LIQUID MIXTURES

- There are different methods in this process: FILTRATION, DECANTATION and CENTRIFUGING.

**A] FILTRATION**

- Occurs when the solute is insoluble. The mixture is poured onto filter paper. The insoluble solute remains on the filter paper and is called a RESIDUE. The liquid passes through the porous filter paper and collects in the container and is called a FILTRATE.

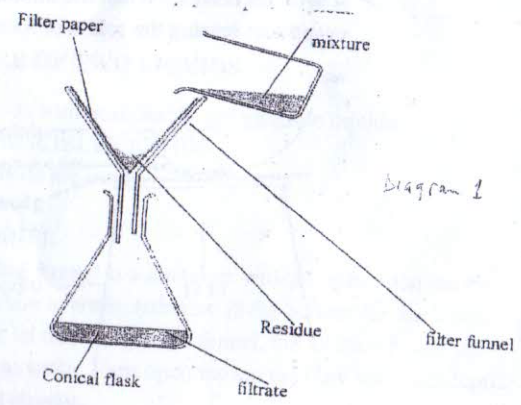



Diagram 1

**B] DECANTATION**

- In this method the insoluble solid is allowed to settle at the bottom of the container. Then the liquid is carefully poured out.

C] **CENTRIFUGING**

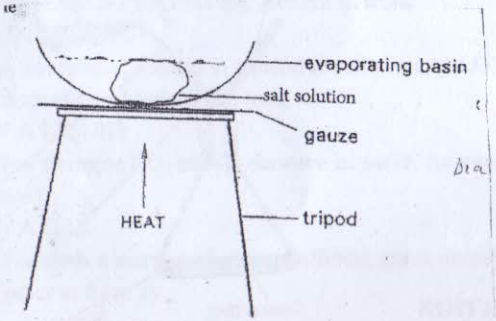
☞ This occurs in a suspension when the solid particles are too small to settle at the bottom of the container. To separate such a mixture, the mixture is spun around very fast and the solid is flung to the bottom of a ***centrifuge tube***, then ***decantation*** follows.

**SEPARATING SOLUBLE SOLID-LIQUID MIXTURE**

➢ Two methods are used: EVAPORATION and CRYSTALIZATION AND DISTILLATION [simple]

1. EVAPORATION AND CRYSTALIZATION

➢ It involves heating. When the mixture is heated, the solvent evaporates leaving the solute in form of grains [crystals].




2. SIMPLE DISTILLATION



➢ In simple distillation, the mixture is heated. The solvent boils and vapour rises. Then vapour condenses in the ***lieberg condenser*** and collects as a DISTILLATE in the flask. The solute remains in the boiling tube. This method is used when both the solute and solvent need to be collected.

Simple distillation

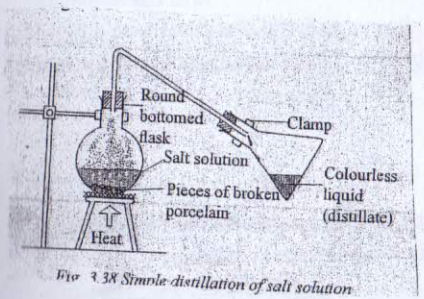



*Fig. 3.38 Simple distillation of salt solution*

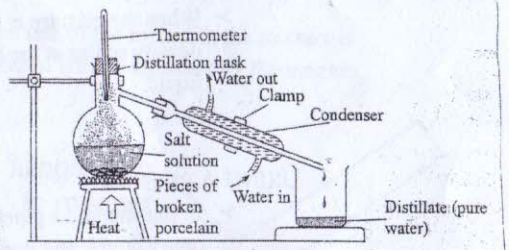



*Fig. 3.39 Simple distillation of salt solution*

**SEPARATING A MIXTURE OF TWO LIQUIDS**

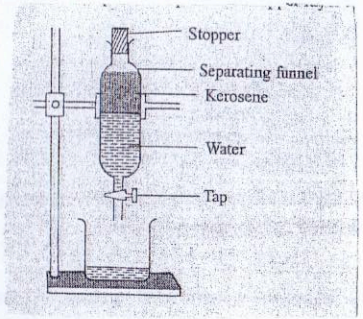
➢ This involves both immiscible and miscible liquids.

A] SEPARATING IMMISCIBLE LIQUIDS

➢ Three methods are used:

1. SEPARATING FUNNEL

➢ A **separating funnel** is a container with an opening at the top for the mixture to enter, and a tap at the bottom for the lower liquid to be let out. To use this funnel, the mixture inside the funnel has to settle. Then open the tap to allow the lower liquid to drain out slowly.



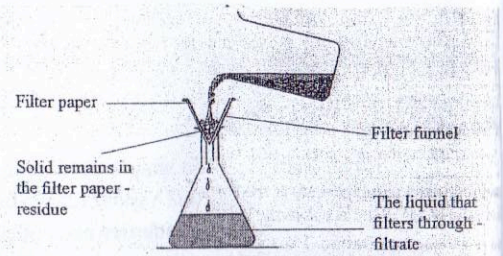




*Fig. 3.34 Separation of soil from water by filtration*

2. DECANTATION

- When the mixture is allowed to settle, the liquid with a higher density settles at the bottom. Then carefully pour out the upper liquid

3. USING A PIPETTE

- In this method a pipette or a dropper is used to suck up the upper liquid; and drain it into another container.

**B] SEPARATING MISCIBLE LIQUIDS**

- In this process, FRACTIONAL DISTILLATION method is used; which uses a FRACTIONATING COLUMN. This separation depends on the fact that the liquids mixed together have ***different boiling points***. The liquid with lower boiling point boils off first and the one with higher boiling point boils off last. The first one condenses and collects in the collecting flask as a DISTILLATE.

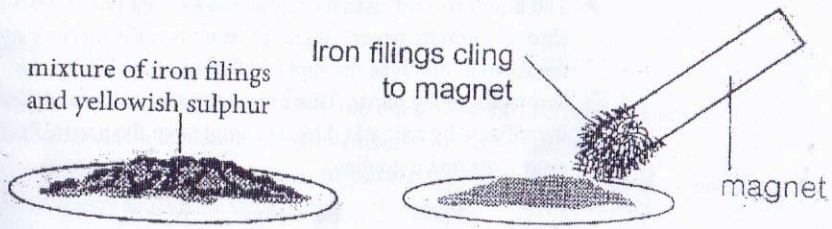


**D] SEPARATING SOLID-SOLID MIXTURES**

This involves:

1. MAGNETISM

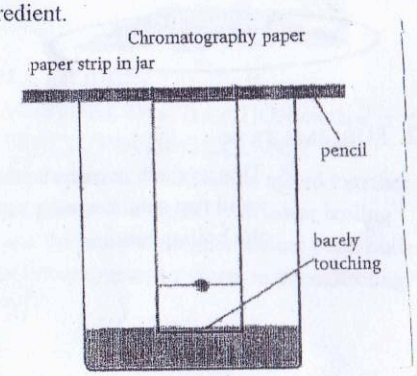
- This method is used when one of the metals in the mixture is **ferrous**. The ferrous material will be attracted by the magnet.




2. SUBLIMATION

- This method involves heating. When the mixture is heated, the solid that sublimes with vapour rises leaving the other one in the boiling tube.

3. CHROMATOGRAPHY
   - ➢ It is a method used to separate a mixture of chemically similar ingredients.
   - ➢ This is used to separate solids or dyes which are soluble in a particular solvent; for example, dyes and inks.
   - ➢ Chromatography paper is normally used to analyse complex mixtures.
   - ➢ The liquid solvent rises by capillary action up the chromatography paper. While passing over the mixture spot, the solvent dissolves the spot and the mixture rises up the chromatography paper. This causes the separation of different ingredients by rising to different heights up the paper. Each spot represents one ingredient.




4. SOLVENT EXTRACTION
   - ➢ This method is used to separate a mixture of solids. One of the solids should be soluble in a particular solvent.

- **Chromatography** can be used to:
  - ➢ Identify different dyes and inks.
  - ➢ Test for drugs in the urine of people who take illegal drugs.
- Solvent extraction is used to extract plant dyes and oil from plants.



# CHAPTER 4

## ATOMIC STRUCTURE

Matter is made up of tiny particles called ATOMS. Atoms chemically combine to form MOLECULES. Atoms can either gain or lose electrons to become IONS.

## JOHN DALTON'S THEORY OF ATOMS

He suggested that:

- All atoms of one element are alike and have the same chemical properties.
- Different atoms combine in small numbers to form molecules.
- Atoms are neither created nor destroyed but they just rearrange themselves to form new substances during chemical reactions.

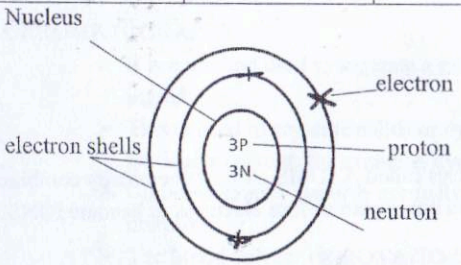
## STRUCTURE OF AN ATOM

- It consists of a nucleus at the centre which contains PROTONS and NEUTRONS. These two are called NUCLEONS because they are found in the nucleus. Around the nucleus are ELECTRONS which revolve around the nucleus at very high velocities in their ELECTRON SHELLS OR ENERGY LEVELS. The electrons are held within the atom by an ELECTROSTATIC FORCE OF ATTRACTION between themselves and the positive charge of the protons in the nucleus.
- The protons, neutrons and electrons are collectively called SUB-ATOMIC PARTICLES.

TABLE

| PARTICLE | POSITION IN ATOM | CHARGE | MASS RELATIVE TO ATOM |
|---|---|---|---|
| Proton | Nucleus | +1 | 1 |
| Neutron | Nucleus | 0 | 1 |

| Electron | Electron shell | -1 | $\frac{1}{1840}$ |
|---|---|---|---|
| | | | |




## ATOMIC NUMBER [Z]

- Atomic number [Z] is equal to the number of protons in the nucleus of an atom. It is the identity of any atom; because no two or more elements have atoms with the same number of protons.

## MASS NUMBER [A]

- Mass number [A] is equal to sum of protons and neutrons in the nucleus. Mathematically,

  A = protons +Z.

## THE GENERAL SYMBOL OF AN ELEMENT

(Mass Number)A←A

X → symbol of element

(Atomic Number)←Z

## SYMBOL, ATOMIC NUMBERS AND MASS NUMBERS OF THE TWENTY ELEMENTS

| ELEMENT | SYMBOL | ATOMIC NUMBERS | MASS NUMBERS |
|---|---|---|---|



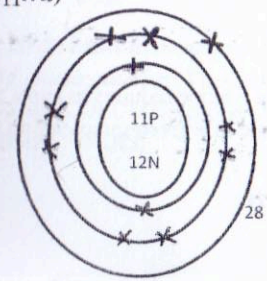
| Hydrogen | H | 1 | 1 |
|---|---|---|---|
| Helium | He | 2 | 4 |
| Lithium | Li | 3 | 7 |
| Beryllium | Be | 4 | 9 |
| Boron | B | 5 | 11 |
| Carbon | C | 6 | 12 |
| Nitrogen | N | 7 | 14 |
| Oxygen | O | 8 | 16 |
| Fluorine | F | 9 | 19 |
| Neon | Ne | 10 | 20 |
| Sodium | Na | 11 | 23 |
| Magnesium | Mg | 12 | 24 |
| Aluminium | Al | 13 | 27 |
| Silicon | Si | 14 | 28 |
| Phosphorous | P | 15 | 31 |
| Sulphur | S | 16 | 32 |
| Chlorine | Cl | 17 | 35 |
| Argon | Ar | 18 | 40 |
| Potassium | K | 19 | 39 |
| Calcium | Ca | 20 | 40 |
| | | | |

## ELECTRON ARRANGEMENT IN ATOMS

- The first electron shell has lowest energy and can only accommodate two electrons.
- The first shell must be filled first before any electron can go to the next one.
- Tradition has it that only shells of an atom with electrons are shown when drawing the structure of an atom.

**Example**

Sodium atom $(^{23}_{11}Na)$



## ELECTRON CONFIGURATION

- It refers to the arrangement of electrons in an atom. It shows the number of electrons in each shell.

**Example**

Na = 11 electrons: 2, 8, 1.

P = 15 electrons: 2, 8, 5.

Ca = 20 electrons 2, 8, 8, 2.

## ISOTOPES

- Isotopes are atoms of the same element having different mass numbers.

*Examples*

a. $^{35}_{17}Cl$ and $^{37}_{17}Cl$

b. $^{12}_{6}C$, $^{13}_{6}C$ and $^{14}_{6}C$

c. $^{235}_{92}K$ and $^{238}_{92}K$

d. $^{1}_{1}H$ ; $^{2}_{1}H$ ; and $^{3}_{1}H$
(protium; denterium and tritium)

## AVERAGE MASS OF ISOTOPES

- To calculate the average mass of an isotope requires to know the percentage or relative abundance of each isotope in a given sample of an element.



***Worked example***

*A sample of chlorine gas contains two isotopes: cl – 35 and cl- 37. 75% of the gas is cl – 35 and 25% is cl – 37. Calculate the relative atomic mass of chlorine.*

***Solution***

$$\text{Average mass} = \frac{mass \times \%\ of\ isotope1 + mass \times \%\ of\ isotope2}{100}$$

$$= \left(\frac{35 \times 75) + (37 \times 25)}{100}\right.$$

$$= \frac{2625 + 925}{100}$$

$$= \frac{3550}{100}$$

$$= \underline{\mathbf{35.5.ans}}$$

☞ **Isotopes** of the same element have similar chemical properties because they have the same atomic numbers.

# CHAPTER 5

## THE PERIODIC TABLE

- The first 20 elements from chapter 4 have been arranged in order of increasing ***atomic number***. These elements are arranged in ROWS called *PERIODS* and COLUMNS called *GROUPS*. This table of elements is called the PERIODIC TABLE.

- Groups are also called **FAMILIES** and periods are also called **SERIES**.

**vm&mm**

PERIODIC TABLE

| | Group I | Group II | Group III | Group IV | Group V | Group VI | Group VII | Group VIII |
|---|---|---|---|---|---|---|---|---|
| Period 1 | H 1 | | | | | | | He 2 |
| Period 2 | Li 2.1 | Be 2.2 | B 2.3 | C 2.4 | N 2.5 | O 2.6 | F 2.7 | Ne 2.8 |
| Period 3 | Na 2.8.1 | Mg 2.8.2 | Al 2.8.3 | Si 2.8.4 | P 2.8.5 | S 2.8.6 | Cl 2.8.7 | Ar 2.8.8 |
| Period 4 | K 2.8.8.1 | Ca 2.8.8.2 | | | | | | |

**ELECTRON CONFIGURATION AND THE PERIODIC TABLE**

Consider sulphur with atomic number 16. Its electron configuration is 2, 8, 6.

- Last figure of this configuration is 6 and sulphur is in group 6.
- This configuration has 3 digits and sulphur is in period 3.
  - Hence the electron configuration of an element shows the **group** and the **period** where the element belongs.

**FAMILIES IN THE PERIODIC TABLE**

- Metals are found on the left and non-metals on right of the table. Lying between the dividing lines are **METALLOIDS** which have characteristics between metals and non-metals.



- Group 1 elements are called ALKALI METALS; group 2 elements are called ALKALINE EARTH METALS, group 7 elements are called HALOGENS and group 8 elements are called INERT OR NOBLE GASES.
- **Alkali metals** are soft and very reactive with oxygen and water hence they are stored in paraffin.
- **Alkali earth metals** are more abundant in earth's crust and just like alkali metals form basic solutions when they react with water.
- **Halogens** are most reactive non-metals. They react with metals to form metal **hallides** like NaCl.
- **Noble gases** are very unreactive gases because their outer shells are filled with electrons.

## CHAPTER 6

### PHYSICAL AND CHEMICAL CHANGES

- Physical changes are temporary e.g. from liquid to solid. Chemical changes are permanent, e.g. formation of water from oxygen gas and hydrogen gas.

**DIFFERENCES BETWEEN PHYSICAL AND CHEMICAL CHANGES**

- All the changes of the states of matter are physical changes. These include shape, size, colour, boiling, melting etc.
- Chemical changes result in the formation of new substances.
- Physical changes involve the breakdown of intermolecular forces only. There is no rearrangement of atoms.
- In chemical changes atoms are rearranged so that new substances are formed.

**CHEMICAL REACTIONS**

- A chemical reaction is the rearrangement of atoms to form new substances. For example, sodium reacts with water to form sodium hydroxide and hydrogen gas i.e.

  $Na + H_2O \rightarrow NaOH + H_2$

- Substances that take part in a chemical reaction are called **REACTANTS** and substances produced are called **PRODUCTS**.
- Chemical reactions are usually represented in form of equations.
- Given as:
  **Reactants → Products**

## CHEMICAL EQUATIONS

➢ There are three different types of reaction equations:

a. **Word equations**

These equations are represented using words

e.g. Carbon + Oxygen → carbon dioxide

Hydrogen + Oxygen → water

b. **Diagram equations**

This is done by representing the reactants and products with diagrams.

e.g. let carbon atom be • and let oxygen atom be ○

carbon + oxygen → carbon dioxide

• + ○○ → ○•○

c. **Chemical symbol equations**

e.g. hydrogen + oxygen → water

$2H_2\ (g) + O_2\ (g) \rightarrow 2H_2O\ (l)$

## RULES FOR WRITING CHEMICAL EQUATIONS

- The reaction should be an experimental fact.
- The formula for reactants and products must be correct.
- Balance number of atoms on both sides of the equation.
- Write the physical state in the formulae(solid [s], gas[g], liquid[l], aqueous solution[aq] .
- Do not change the formula of reactants and products.

## COEFICIENT

- It is the number written in front of a molecular formula ($6H_2O$ – 6 is the coefficient).



- It shows the number of molecules of the chemical formula.

## SUBSCRIPT

- It is the number written slightly below the symbol of the element or atom.
- It indicates the number of atoms of the element ($6H_2O$ – 2 is the subscript).

## SOME ATOMIC SYMBOLS AND MOLECULAR FORMULAE

| TYPE | SUBSTANCE | ATOMIC SYMBOL | MOLECULAR FORMULA |
|---|---|---|---|
| Diatomic element | Oxygen | O | $O_2$ |
| | Nitrogen | N | $N_2$ |
| | Bromine | H | $H_2$ |
| | Chlorine | Cl | $Cl_2$ |
| | | | |
| Mono-atomic elements | Carbon | C | - |
| | Magnesium | Mg | - |
| | Sodium | Na | - |
| | Zinc | Zn | - |
| | Potassium | K | - |
| | Calcium | Ca | - |
| | | | |
| Compounds | Water | H and O | $H_2O$ |
| | Carbon dioxide | C and O | $CO_2$ |
| | Ammonia | N and H | $NH_3$ |
| | Calcium chloride | Ca and Cl | CaCl |
| | Potassium bromide | K and Br | KBr |
| | Nitrogen dioxide | N and O | $NO_2$ |

**For example**

Hydrogen + Oxygen → water

$2H_2\ (g) + O_2\ (g) \rightarrow 2H_2O\ (l)$

## BALANCING CHEMICAL REACTION EQUATIONS

***Guidelines***

1. Total number of reacting atoms must be equal to the number of atoms in the products.
2. The kinds of atoms involved in the reaction do not decrease or increase but remain the same.

***There are two methods of balancing chemical equations***

- *Trial and error method*
- *Systematic (algebraic) method*

**For example**

Balance the following equation $N_2 + H_2 \rightarrow NH_3$

➢ $N_2 + 3H_2 \rightarrow 2NH_3$

## REACTING MASSES

➢ Amount of products can be calculated from the given amount of reactants. Likewise, amount of reactants can be worked out from the given amount of products.

## CALCULATING RELATIVE ATOMIC MASS (RAM)

➢ It refers to the ratio of the mass of 1 atom in a.m.u to the $\frac{1}{12}$ of the mass of an atom of carbon.

Mathematically

$$RAM = \frac{mass\ of\ 1\ atom\ of\ an\ element\ in\ amu}{\frac{1}{12}\ of\ mass\ of\ 1\ atom\ of\ Carbon}$$

***Worked Example***

What is the RAM of potassium atom?



***Solution***

Mass of K = 39amu

Mass of C = 12amu

$$RAM\ of\ K = \frac{39amu}{\frac{1}{12}\times 12amu} = \frac{39amu}{1amu}$$

$$= \underline{39\ Ans.}$$

☞ This means that **RAM** has no units.

## CALCULATING RELATIVE FORMULA MASS (RFM)

➢ **RFM** is also known as **Relative Molecular Mass (RMM).**

➢ **RFM** is found when there is more than 1 atom making a molecule of a substance.

➢ This is calculated by adding the **RAMs** of the atoms in the formula.

***Worked Example***

*Calculate the RFM of MgO.*

***Solution***

*RAM of Mg = 24*

*RAM of O = 16*

*RFM of MgO = 24 + 16 = **40 ans.***

## CALCULATING REACTING MASSES

➢ Mass of a reactant can be determined by mass of another reactant.

***Worked example***

Magnesium reacts with oxygen to produce magnesium oxide according to the equation

$2Mg\ (s) + O_2\ (g) \rightarrow 2MgO\ (s)$.

If 12 grams of Mg were used, find the mass of:

a. Oxygen used.
b. MgO produced.

*Solution*

$2Mg\ (s) + O_2\ (g) \rightarrow 2MgO\ (s)$

$(2\times24) \quad (2\times16) \quad 2(24+16)$

48g 32g 80g

a. If 48g of Mg is used, 32g of oxygen is used.

Therefore, 12g of Mg $= \frac{12g}{48g} \times 32g =$ <u>8g of O used.</u>

b. **48g** of Mg produces **80g** of MgO

Therefore, 12g of mg will produce ? of MgO

$\frac{12g}{48g} \times 80g =$ <u>20g of MgO produced</u>.

## PERCENTAGE COMPOSITION BY MASS

- This means the percentage of an element in a compound.

*Mathematically*,

$$\text{Percentage of X} = \frac{mass\ of\ the\ element \times 100}{mass\ of\ the\ compound\ (RFM)}$$

*Worked example*

Work out the percentage composition of nitrogen in Ammonium nitrate ($NH_4NO_3$)

*Solution*

Mass of N = 2N = 2×14 = 28g

RFM of $NH_4NO_3$ = 2N + 4H + 3(3O)

= 2(14) + 4(1) + 3(16)

= 28 + 4 + 48 = 80g



$\therefore$ % of N $= \frac{28g}{80g} \times 100 =$ ***<u>35% ans</u>***

# CHAPTER 7

## ORGANIC COMPOUNDS

- Organic compounds are substances that contain carbon-hydrogen bonds in their formulae. Compounds with carbon and without hydrogen atoms are INORGANIC.
- Examples of organic compounds are ALKANES, ALKENES, ALKANOLS and CARBOXYLIC ACIDS.

## NATURAL SOURCES OF ORGANIC COMPOUNDS

1. **Animals**: since they contain carbohydrates, [C, H, O] Lipids [C, H, O] and Proteins [C, H, O, N, S].
2. **Plants:** these contain same materials as animals.
3. **Fossils**: these are the remains of dead organisms [i.e. plants and animals].

## USES OF ORGANIC COMPOUNDS AS FUELS

- A fuel is any substance which reacts with oxygen to produce $CO_2$, $H_2O$ and Heat energy. Examples are coal, charcoal, petrol, paraffin, diesel, ethanol etc.

## ORGANIC COMPOUNDS AS FOSSIL FUELS

1. **Coal**: this is a solid fossil fuel which is the remains of trees, shrubs and grasses that lived as much as two million years ago.
2. **Petroleum**: known as crude oil. It is formed from remains of tiny plants and animals that lived in the sea millions of years ago.

3. **Natural gas**: a fossil fuel which is formed from layers of buried plants and animals after having been exposed to heat and pressure over thousands of years. Examples are **alkanes** of 1 - 4 carbon atoms.

## ORGANIC COMPOUNDS AS BIO-FUELS

- Bio-fuels are fuels that contain energy from geologically recent carbon fixation such as plants and animals. Examples include Biogas, ethanol and biodiesel.

A] BIOGASS: It is produced from the breaking down of organic matter in the absence of oxygen. Bacteria breakdown animal dung and human waste put in DIGESTER without oxygen. As a result, methane gas is produced which is collected and used to produce heat.

**Advantages of biogas are:**

- Non- polluting and renewable source of energy.
- Remains are used as manure.
- Reduces deforestation.
- Time saved for collecting wood and cooking.

B] BIODIESEL

- It is produced from breaking down of vegetable oil and animal fats.

**Advantages**

- Less pollutant.
- Can be blended with other energy resources and oil.
- It has lubricating property which may increase the life span of engines.

C] BIOETHANOL



- This can be used in petrol engines. It is produced from alcohol.

**Advantages**

- It reduces certain greenhouse emissions.
- It is cheap because it is produced from locally available sugarcane.
- It is easy to use.

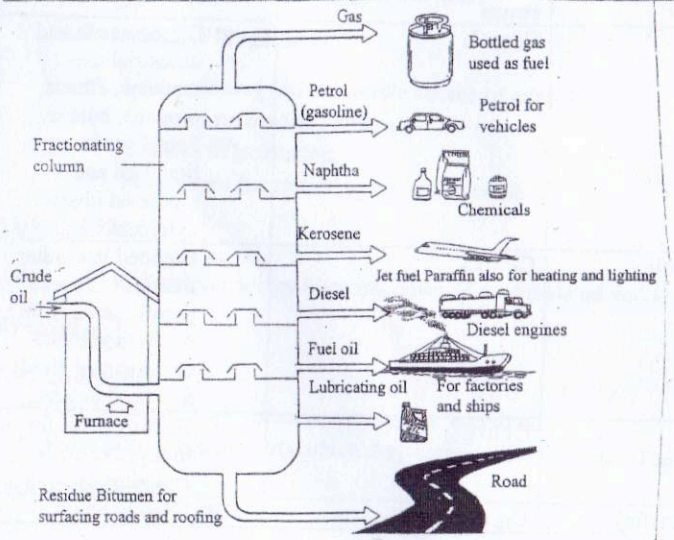
## PETROLEUM AND ITS COMPONENTS

- Petroleum is a mixture of HYDROCARBONS. The individual components are extracted through FRACTIONAL DISTILLATION because are fractions of the whole lot. This is based on the fact that different fractions have different boiling points.

## USES OF FRACTIONS OF PETROLEUM

| Name of fraction | Number of carbon atoms | Uses |
|---|---|---|
| Petroleum gas | $C_1$ to $C_4$ | For domestic and industrial use- methane, ethane, propane, butane. Petrol gas is liquefied and marketed in steel cylinders. |
| Gasoline [petrol] | $C_5$ to $C_6$ | Blended into other fractions to make petrol used as fuel for light vehicles, dry-cleaning fluid and used in labs. |
| Naphthalene | $C_7$ to $C_{10}$ | Starting point for many chemicals and plastics. |
| Kerosene [paraffin] | $C_{10}$ to $C_{16}$ | Used to jet oil |

| | | |
|---|---|---|
| | | ☞ Used as detergent<br>☞ Used domestically |
| Diesel oil | $C_{17}$ to $C_{20}$ | ☞ Fuel for large vehicles |
| Lubricating oil | $C_{21}$ to $C_{30}$ | ☞ Oils for cars and other machines |
| Fuel oil | $C_{31}$ to $C_{40}$ | ☞ Fuels for power station, ships, heavy vehicles and industrial engines. |
| Paraffin waxes | $C_{41}$ to $C_{50}$ | ☞ Manufacturing of candle wax, polish, waxed paper, water-proofing, grease, Vaseline, ointments, cream cosmetics etc. |
| Bitumen | $C_{51}$ upwards | ☞ Pitch for road and roofs and damp proofing foundation of the buildings. |






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# FORM 2

## TOPICS

1. Elements and the periodic table
2. Chemical bonding
3. Acids and bases
4. Organic Chemistry
5. Air
6. soil

# CHAPTER 1

## ELEMENTS AND THE PERIODIC TABLE

- Elements in the periodic table are ***metals*** [on the left], ***metalloids*** [in the middle] and ***non-metals*** on the right.

## PERIODIC TRENDS

- A trend is a behavioral pattern that may occur repeatedly. Among the patterns in the periodic table are ***atomic radius, electron affinity, electronegativity*** and ***ionization energy***.

### 1. ATOMIC RADII

- The radius of an atom is the distance from the centre of the atom to the farthest shell.
- Across the periodic table atomic radii of elements decrease. This is because of increased attraction between nucleus and outer electrons as the atomic number increases and electrons are being added to the same shell.



- Down the group the atomic radii increase due to decrease in attraction between nucleus and outer electrons because an extra shell is being added and the shells cause shielding of nuclear attraction.

### 2. IONISATION ENERGY

- Ionization energy is the minimum energy required to remove electrons from an atom.
- Down the group ionization energy decreases because electrons in the enter shell are less attracted to the nucleus.
- Across the period ionization energy increase because electrons in the outer shell increase and experience more attraction force.

### 3. ELECTRONEGATIVITY

- This is the ability of an atom to attract electrons to itself.
- Across the period, electronegativity increases due to an increase in nuclear charge as the atomic number increases.
- Down the group, the electronegativity decreases due to increase in radius as the number of shells increases.

### 4. ELECTRON AFFINITY

- This is the ability of an element to accept electrons. This is the opposite of electronegativity.
- Effective nuclear charge [force of attraction] increases across the period due to increase in atomic number.
- Electron affinities, therefore, increase across the periodic but decrease down the group due to increase in atomic radii.

### 5. VALENCE ELECTRONS

- These are the outer electrons. Across the period, valence electrons increase and down the group, valence electrons remain the same.

### 6. METALLIC CHARACTERISTICS

- These characteristics decrease from left to right of the periodic table.

PROPERTIES OF SOME GROUPS

A] ALKALI METALS

*1] PHYSICAL PROPERTIES*

- They are soft [potassium is the softest].
- They are shiny when freshly cut.
- They float on water i.e. they have lower density than water.

*2] CHEMICAL PROPERTIES*

- They react with water to form a hydroxide and hydrogen gas
  e.g. $2\ Na\ (s) + 2H_2O\ (l) \rightarrow 2NaOH\ (aq) + H_2\ (g)$
- They are the most reactive metals.
- They react with oxygen to form an oxide
  $4\ Li\ (s) + O_2\ (g) \rightarrow 2\ LiO\ (s)$
- They react vigorously with halogens producing metal halides in the process
  $2K\ (s) + Cl_2\ (g) \rightarrow 2KCl\ (s)$

- In group 1, **reactivity** increases down the group because atomic radii increases down the group so the valence electrons is weakly attracted to the nucleus and the element can easily become an ion.

USES OF ALKALI METALS

1. **Lithium**
   - ℜ In glasses and ceramics that resist heat.
   - ℜ Its alloys can be used in aircraft buildings.
   - ℜ For making cellphone and computer batteries.
   - ℜ For making lubricating greases.
   - ℜ Its salts are used for making drugs for stabilizing mood.



2. **Sodium**
   - ℜ As coolants in nuclear reactors.
   - ℜ As a luster in metals.
   - ℜ Its compounds are used in petroleum, textile and paper industries.
   - ℜ In street lights and sodium vapour lamps.
   - ℜ Its hydroxide is used in soap making.

3. **Potassium**
   - ℜ Its nitrates are used for making explosives, fireworks and food preservatives.
   - ℜ Its chromate is used for tanning leather and manufacture of inks, gun powder, dyes and safety matches.
   - ℜ Its hydroxide is used for making soap.

**B] ALKALINE EARTH METALS**

**USES**

1. Beryllium is used in high strength electrical insulators and semi-conductors.
2. Magnesium is mixed with Aluminium or Zinc to alloys of more desirable qualities.
3. Calcium is used in separating metals from their ores.

**C] THE HALOGENS**

1. PHYSICAL PROPERTIES

- ℜ Are made up of DIATOMIC molecules [$F_2$, $Cl_2$ etc].
- ℜ Have low melting and boiling points.
- ℜ Are poisonous and have different colours.
  Fluorine –pale yellow gas
  Chlorine –greenish yellow gas
  Bromine –brown liquid
  Iodine –blue, black shiny solid

ℜ Atomic radii increases down the group.

2. CHEMICAL PROPERTIES

They are very reactive non-metals.

ℜ They react with hydrogen to produce hydrogen halides
$H_2$ [g] + $Br_2$ [l] → 2HBr (l).

ℜ They react with alkali metals to form metal halides.
2Na (s) + $Cl_2$ (g) → 2NaCl (s)

ℜ Reactivity decreases down the groups because the atomic radii increases, hence smaller atoms can easily become ions.

## USES OF HALOGENS

- Fluorine is used in the form of fluorides in drinking water and toothpaste to reduce tooth decay.
- Chlorine is used to make plastic called polyvinyl-chloride [PVC] for electrical insulators and making household bleaches e.g. harpic.
- Chlorine can be used to kill bacteria and viruses in drinking water.
- Bromine is used in photography as silver bromide [Ag Br] and medicine as potassium bromide [KBr].
- Bromine is used as flame retardant in fire extinguishers.

## D] INERT [NOBLE] GASES

- These do not take part in chemical reactions [they are unreactive] because they fill outermost shells.

## PHYSICAL PROPERTIES

- They are colourless gases.
- They are mono-atomic elements.



## USES

- Argon is used to fill the bulbs to prevent the tungsten filament from reacting with air.
- Neon:
  A] Glows when electricity passes through it, hence used in advertising signs.
  B] Used in Geiger-Muller [Gm] tubes to detect radiation.
  C] Together with Helium used in helium-neon laser gas that is used in eye surgery.

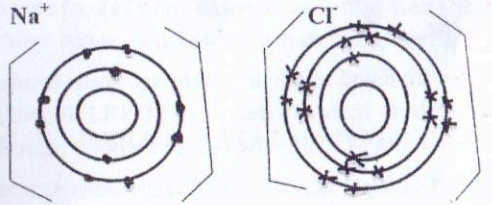
**3. Helium**

- Used as a coolant in nuclear reactors.
- Used to inflate tyres of large aircraft to make them light enough.
- Used to fill airships and weather balloons to make them very light and cause them float on water and in air.

# CHAPTER 2
## CHEMICAL BONDING

A] IONIC [ELECTROVALENT] BONDS

- These are formed when atoms either lose or gain atoms i.e. become ions. Positive ions are **CATIONS** and negative ions are **ANIONS**.
- The ***cations*** and ***anions*** attract to form a bond. For example, $Na^+$ and $Cl^-$.




IONIC LATTICE

- This is an ionic CRYSTAL [grain]. Ions in an ionic lattice are arranged side by side.

## FORMULAE OF IONIC COMPOUNDS

- The easier method of writing formulae of ionic compounds is to swap the charges between the ions and write them as **subscripts of the atomic symbols** leaving out signs. For example, the formula of magnesium chloride:

  Magnesium is $Mg^{2+}$, Chloride is $Cl^-$, hence the formula is:

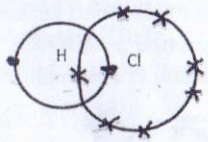
☞ $Mg^{2+}Cl^- = MgCl_2$

## PHYSICAL PROPERTIES OF IONIC COMPOUNDS

- They have high melting and boiling points.
- They conduct electricity in their molten state because they have free ions.
- They are soluble in water to from aqueous solutions.

## B) COVALENT (MOLECULAR ) BONDS

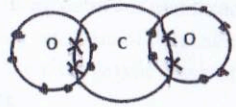
- These bonds are formed by sharing valence electrons.
  e.g. hydrogen chloride (HCl)




- The shared electrons are found where the outer-most shells overlap.
- When one pair of electrons is shared, then it is a SINGLE BOND, when 2 pairs are shared, then it is a DOUBLE BOND, and when 3 pairs are shared, then it is a TRIPLE BOND.

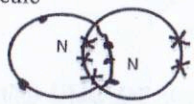
**Example**
Carbon dioxide





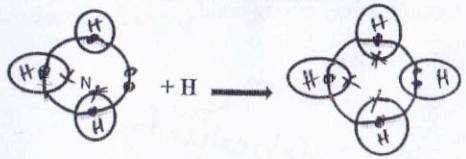

**double bond**

In Nitrogen molecule
**triple bond**




## DATIVE (CO – ORDINATE) BOND

- This is a covalent bond in which the shared pair of electrons is provided by one of the bonded atoms.
- This pair of electrons is shared with an electron that has lost its electron(s) in the outer shell and has become ionized. The atom providing electrons is called a DONOR and the one receiving is called an ACCEPTOR.
- A dative bond is formed in ammonium ion.




**Dative bond**

## MOLECULAR COMPOUNDS

- They are formed when different non-metal atoms share electrons to form covalent bonds, such as hydrogen chloride, methane, glucose etc.

## FORMULAE OF SIMPLE MOLECULAR STRUCTURES

- This is possible when the valence electrons are known. For example, the molecular formula of methane ($CH_4$) is:

$C^4 H_4^1 \rightarrow CH_4$

Ammonium is:

$$N\cancel{H_3} \rightarrow NH_3$$

## PHYSICAL PROPERTIES OF MOLECULAR SUBSTANCES

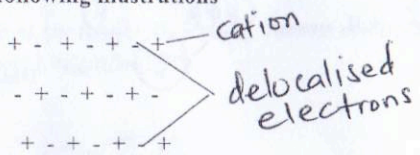
- They have low melting and boiling points.
- They do not conduct electricity.
- They are insoluble in water but soluble in organic compounds.

### D] METALLIC BONDS

These bonds are made between the stationary positive ions and free delocalized electrons. There is attraction between these ions.

These are the strongest bonds because each free delocalized electron is attracted in there directions, hence forming a very strong bond.

**Consider the following illustrations**




## STRENGTH OF METALLIC BOND

**Occurs mainly due to:**

### A] SIZE OF METAL ATOMS

- The electrostatic force decreases going down the group. Hence substances from the upper part of the group will be having stronger electrostatic forces and they will be hard.

### B] NUMBER OF VALENCE ELECTRONS CONTRIBUTED TO THE 'SEA' OF ELECTRONS



- The melting and boiling points of metals across the periodic increase because of increasing numbers of delocalized electrons. As the delocalized electrons increase, the electrostatic force of attraction between the electrons and nuclei increases.

## PROPERTIES AND USES OF METALS

- They have high melting and boiling points; because they have very strong bonds.
- They expand when heated.
- Metals are good conductors of heat and electricity.
- They are malleable [can be beaten into various useful shapes e.g. sheets].
- Metals are ductile [they can be drawn into wires].
- They are sonorous [they produce a ringing sound].
- Metals are dense and strong.

# CHAPTER 3

## ACIDS AND BASES

- Acids have a sour taste while bases have a flat taste
- Both acids and bases conduct electricity [they are electrolytes]
- Both acids and bases react with metals

## ACID/BASE INDICATORS

- Indicators are used to distinguish acids from bases.
- Indicators change different colours when put in acids and bases.
- Examples of indicators are universal indicator, bromothymol blue, phenolphthalein and litmus.

## PREPARING ACID/BASE INDICATORS FROM LOCAL MATERIALS

- Indicators are usually dyes extracted from plants. The dyes are found in plant leaves and flowers.

## STRENGTH OF ACIDS AND BASES [pH scale]

- The strength of an acid or a base is determined by using ELECTRICAL CONDUCTIVITY, a UNIVERSAL INDICATOR or a pH METER.
- The strength of a base or acid can be known by using a pH scale. PH stands for power of hydrogen. The pH scale has corresponding colours as can be seen below:

| COLOUR | pH SCALE | |
|---|---|---|
| Red | 1 | |
| Pink | 2 | |
| Orange | 3 | |
| Orange | 4 | |
| Orange-yellow | 5 | increasing acidity |
| Yellow | 6 | |
| Greenish-yellow | 7 | neutral |
| Green | 8 | |
| Green-blue | 9 | increasing alkalinity |
| Blue | 10 | |
| Violet | 11 | |
| Purple | 12/13/14 | |

**pH SCALE**

| Ph | 0&1 | 2 | 3&4 | 5 | 6 | 7 | 8 | 9 and 10 | 11 | 12/13/14 |
|---|---|---|---|---|---|---|---|---|---|---|
| Colour | Dark red | Red | orange | Yellow | Light green | Green | Dark green | blue | Dark blue | Purple |

- pH 0 – 6 means the solution is ***acidic***.



- pH 7 means the solution is ***neutral***.
- pH 8 – 14 means the solution is alkaline or basic.
  - If the pH value is very small then the solution is very **acidic**.
  - If the pH value is very great then the solution is very **alkaline** or **basic**.

**Why is a universal indicator preferred when measuring pH value of a solution?**

- It provides a wide range of colour.

**Examples of indicators and their colours**

| Name of indicator | Basic colour | Acidic colour |
|---|---|---|
| Litmus paper | Blue | Red |
| Phenolphthalein | Pink | Colourless |
| Bromothymol blue | Blue | Yellow |
| Universal indicator | Blue | Red |

## USES OF ACIDS AND BASES

- Sulphuric acid [$H_2SO_4$] is used in car batteries and manufacturing industries producing. paints, fertilizers and soap less detergents like surf.
- Ethanoic acid [acetic acid] is used as a food preservative and a solvent.
- Methanoic acid is used in dyeing and electroplating.
- Bases are used in food preparation e.g. okra [thelele].
- Bases are used to manufacture soap.
- Bases are used to neutralize acidic soils and acids are used to neutralize alkali soils.

## NEUTRALISATION

- This is a process of reducing acidity or basicity of a substance. All acid-base reactions are neutralization process.

**Neutralization is applied in:**

A] IN DIGESTION: the acidity in the stomach can be reduced by stomach powder such as drew liver salt, magnesium trisilicate and milk of magnesium.

B] USED IN AGRICUTURE: some crops do well in acid soils, others in neutral soils and basic soils. So acidic alkali soils have to be neutralized to accommodate crops which prefer neutral soils

C] USED IN CLEANING TEETH: acids produced by micro-organisms in the mouth are reduced by alkali bases.

## CHAPTER 4

### HYDROCABONS

**Hydrocarbons** are organic compounds which contain carbon and hydrogen atoms only.

**Organic compounds** are substances which contain C-H bond. There are 3 classes of organic compounds, namely:

1. **HDROCARBONS:** substances which contain carbon and hydrogen atoms only; e.g. alkanes [$C_nH_{2+n}$] and alkenes [$C_nH_{2n}$].
2. **OXYCARBONS:** substances which contain carbon, hydrogen and oxygen atoms e.g. alkanols [alcohols] like $CH_3OH$.
3. **NITROCARBONS:** These contain carbon, hydrogen and nitrogen atoms e.g. $CH_3NO_2$.

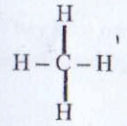
### FAMILIES OF HYDROCARBONS

- There are several families of organic compounds each of which form a **HOMOLOGOUS SERIES** with similar characteristics.
- **Homologous series** are series of compounds related by the same general formula and with similar proportions. The two families or homologous series of hydrocarbons are **alkanes** and **alkenes.**

### ALKANES [PARAFFINS]



- In this homologous series, each carbon atom forms 4 simple covalent bonds with other atoms.



- Alkanes are called SATURATED hydrocarbons because the carbon atoms have 4 single bonds and the carbons atom cannot take in anymore atoms.

### GENERAL FORMULA FOR ALKANES

The general formula is $C_nH_{2n+2}$, when **n** stands for the number of carbons. If n=1, then the formula is $CH_4$, when n=2 the formula is $C_2H_6$ etc.

### NOMENCLATURE OF ALKANES

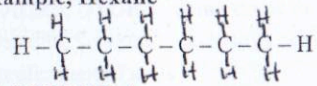
- Nomenclature is a naming system. Members that belong to a particular family have similar names.
- Alkanes are named by adding a suffix ''-ane'' to the common prefixes.The names of the first ten members of alkanes are:

| NUMBER OF CARBONS | COMMON PREFIX | NAME OF ALKANE |
|---|---|---|
| 1 | Meth- | Methane |
| 2 | Eth- | Ethane |
| 3 | Prop- | Propane |
| 4 | But- | Butane |
| 5 | Pent- | Pentane |
| 6 | Hex- | Hexane |
| 7 | Hept- | Heptane |
| 8 | Oct- | Octane |
| 9 | Non- | Nonane |
| 10 | Dec- | Decane |

### STRUCTURAL FORMULA OF ALKANES

➢ A structural formula shows how the atoms are arranged and joined in a molecule.

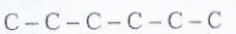
**For example, Hexane**



**SKELETAL FORMULA**

This formula shows the carbon chains i.e. all hydrogens are removed.

**For example, Hexane**



**CONDENSED FORMULA**

This formula shows both the carbon and hydrogen atoms and how many hydrogen atoms per each carbon atom. For example Hexane is $CH_3\ CH_2CH_2\ CH_2CH_2CH_3$ or $CH_3\ (CH_2)_4\ CH_3$

**PHYSICAL PROPERTIES OF ALKANES**

- Alkanes are insoluble in water. they are HYDROPHOBIC [water-hating].
- At room temperature, alkanes with up to 4 carbon atoms are GASES and those with 5. to16 carbon atoms are LIQUIDS and those with more than 16 carbon atoms are SOLIDS.
- Alkanes with larger molecules have higher melting and boiling points because the **longer** the carbon chain the **bigger** the molecular and the greater the **intermolecular forces** [imfs] and the more the heat required to weaken the force. Hence higher ***melting*** and ***boiling*** points.
- The bigger the molecule the greater the density.
- The larger the alkane, the more viscous it is. **Viscosity** is defined as the resistance of a liquid to flow.
- Alkanes do not conduct electricity. They are non-electrolyte.

**CHEMICAL PROPERTIES OF ALKANES**

1. They take part in combustion reaction



**For example**

$CH_4\ (g) + 2O_2\ (g) \rightarrow CO_2 + 2H_2O$ + Heat Energy

2. They take part in substitution reaction in which the hydrogen atoms can be substituted by another atom e.g. a halogen.

**For example**

$CH_4\ (g) + Cl_2\ (g) \rightarrow CH_3Cl + HCl$ [chloromethane]

$CH_3Cl\ (g) + Cl_2\ (g) \rightarrow CH_2Cl_2 + HCl$ [dichloromethane]

$CH_2\ Cl_2 + Cl_2 \rightarrow CHCl_3 + HCl$ [trichloromethane]

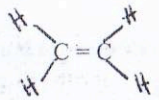
$CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$ [tetrachloromethane or Carbon tetrachloride]

**USES OF ALKANES**

- For heating and cooking e.g. natural gas.
- Alkanes in form of bitumen, are used for surfacing the roads.
- They are used as fuels for vehicles, ships, geoplanes etc.
- They are also used to store the most reactive metals such as lithium, potassium etc.
- They are used as lubricating oil, where moving parts of a machine are lubricated to prevent rusting and reduce friction.
- They are used to produce other organic compounds such as alkenes.
- They are used as the solvent for other organic substances.

**ALKENES**

- Alkenes, unlike alkanes, are unsaturated compounds. The carbon atoms can accept more atoms. They also possess a DOUBLE BOND between a pair of carbon atoms.



- This double bond is called a FUNCTIONAL GROUP. A functional group is the group of atoms or bonds within a molecule that determines the properties of a compound.
- Alkenes are produced from **alkanes** through the process called CATALYTIC CRACKING. This is a process where a long-chain alkane is split up, under high pressure and temperature and in the presence of ***aluminium*** and **chromium** as catalysts.**For example,**

$$C_{10}H_{22} \xrightarrow[Chromium\ oxides]{Aluminium} C_5H_{12} + C_3H_6 + C_2H_4$$

Cracking is subjecting larger hydrocarbon molecules to high pressure at high temperature to break them into smaller molecules in the presence of a catalyst.

$$\textbf{Pentane} \xrightarrow[high\ pressure-high\ temperature]{catalyst} \textbf{propane + ethene}$$

## GENERAL FORMULA OF ALKENES

➢ It is $C_nH_{2n}$, where **n** stands for the number of carbon atoms. The smallest alkene has 2 carbon atoms because of the double bond.

## NONENCLATURE OF ALKENES

➢ The prefix is the same as those of alkanes. The suffix for alkenes is ``**-ene**''. The names of the first 9 alkenes are:

| NUMBER OF CARBONS | COMMON PREFIX | NAME OF A ALKENE |
|---|---|---|
| 2 | Eth- | Ethene ($C_2H_4$) |
| 3 | Prop- | Propene ($C_3H_6$) |
| 4 | But- | Butene ($C_4H_8$) |
| 5 | Pent- | Pentene ($C_5H_{10}$) |
| 6 | Hex- | Hexene ($C_6H_{12}$) |
| 7 | Hept- | Heptene ($C_7H_{14}$) |

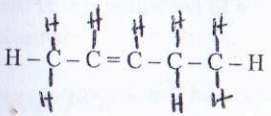


| | | |
|---|---|---|
| 8 | Oct- | Octene ($C_8H_{16}$) |
| 9 | Non- | Nonene ($C_9H_{18}$) |
| 10 | Dec- | Decene ($C_{10}H_{20}$) |

## FORMULAE OF ALKENES

a. STRUCTURAL

For pentene



b. SKELETAL

For pentene

C – C = C – C – C

c. CONDENSED

For pentene

$CH_3CH = CHCH_2CH_3$

☞ The naming of alkenes follows the IUPAC system of naming where the position of the carbon – carbon bond is given.

**Example**

a. C – C = C – C – C (1 2 3 4 5) Pent – 2 – ene or 2 – Pentene.

b. C – C – C – C = C – C – C (7 6 5 4 3 2 1) Hept – 3 – ene or 3 – Heptene.

☞ The numbering of carbon atoms starts closer to the position of the double bond.

## PHYSICAL PROPERTIES OF ALKENES

- They are insoluble in water.
- They exist in all states at room temperature.
- Their melting and boiling points increase as the number of carbon atoms.

- The density of alkenes increases with increases with increase in molecular size.
- Viscosities follow the same trend as other organic compounds.
- They do not conduct electricity.

## CHEMICAL PROPERTIES OF ALKENES

➢ They are more **reactive** than alkanes because of the presence of the **double bond.** They take part in combustion and addition reactions:

### A] COMBUTION REACTIONS.

- They react with oxygen to produce $CO_2$ , $H_2O$ and Heat Energy.

**For example**

$C_2H_4\ (g) + 3O_2\ (g) \rightarrow 2CO_2\ (g) + 2H_2O\ (g) + Heat$

### B] ADDITION REACTIONS.

➢ In addition reactions, two or more molecules combine to form a larger molecule.

1. **HALOGENATION REACTION**

- Halogens are added onto the alkenes.

**For example**

$CH_2 = CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$ [bromoethane]

2. **CATALYTIC HYDROGENATION**

- Hydrogen atoms are added to alkenes.

**For example**

$CH_2 = CH_2 + H_2 \rightarrow C_2H_6$ [ethane].

☞ The reaction takes place under very high temperature.

3. **HYDRATION**

- Water is added to an alkene.

$CH_2 = CH_2 + H_2O \rightarrow C_2H_5OH$ [ethanol].



## USES OF ALKENES

- Used to produce alkanols at industrial level; when they undergo process of hydration.
- Polymerization alkenes are monomers. The monomer react to produce polymers e.g. in the formation of polythene [many ethene molecules].
- Some are used as plant hormones that regulate fruit ripening, flower maturity and seed germination e.g. ethene.
- Used in the production of halogen alkanes which are important industrial solvents.

# CHAPTER 5

## AIR

➢ Air is composed of many kinds of gases. The gases are found in different percentages.

| COMPONENT | % BY VOLUME |
|---|---|
| Nitrogen | 78.08 |
| Oxygen | 20.95 |
| Argon | 0.93 |
| Carbon dioxide | 0.03 |
| Neon | 0.002 |
| Helium | 0.0005 |
| Methane | 0.00017 |
| Krypton | 0.000114 |
| Trace gases | 0.000056 |

➢ In addition to these gases, air also contains water vapour, dust particles, smoke and ash.

## SEPARATION OF COMPONENTS OF AIR

- Air is a mixture of gases.

- **Fractional distillation** is the best method used to obtain individual gases from air. This is possible because the gases have ***different boiling points***. This means that the gases must condense first.

| GAS | BOILING POINT [$^0$C] |
|---|---|
| Xenon | -108 |
| Krypton | -157 |
| Oxygen | -183 |
| Argon | -186 |
| Nitrogen | -196 |
| Neon | -246 |
| Helium | -269 |

## PORPERTIES AND USES OF GASES

### A] NITROGEN GAS

- Nitrogen is a diatomic molecule where the two atoms are bonded by a triple bond [N≡N].
- The triple bond is very strong and cannot easily be broken. This property makes nitrogen to be a very unreactive gas.

### USES OF NITROGEN GAS

1. **Food packaging:** nitrogen prevents oil from reacting with oxygen and the food stays for a longer period.
2. **Fire extinguishing:** nitrogen stops oxygen from reaching the fire when it is spread onto the fire from the extinguisher.
3. **Freezing:** liquid nitrogen is used in refrigeration because at low temperature liquid $N_2$ can freeze foods quickly.
4. **Manufacturing of Ammonia gas:** this is done by the HABER PROCESS. Ammonia gas has several uses.
5. **Preventing fires in oil tankers:** when $N_2$ is put in fuel tankers it provides an inert atmosphere which prevents fires.



### OXYGEN GAS [$O_2$]

- This is a very reactive gas. It reacts with most compounds to substances called OXIDES. The reaction with oxygen is called a COMBUSITION REACTION.

### REACTION OF OXYGEN WITH METALS

- This produces metal oxides.

**For example**

$4Li\ (s) + O_2\ (g) \rightarrow 2Li_2O$ (Lithium Oxide)

$Mg\ (s) + O_2\ (g) \rightarrow MgO$ (magnesium oxide)

$4Al\ (s) + 3O_2\ (g) \rightarrow 2Al_2O_3$ (Aluminium oxide)

### THE REACTIVITY SERIES FOR OXYGEN

- This depends on the rate of reaction between oxygen and different metals. Some metals react quickly and others react slowly. For example:

K Na Ca Mg Al Zn Fe Sn Pb Cu Ag Au

Decreasing Reactivity

### TEST FOR OXYGEN

- The test for oxygen is that it can **relight** a burning splint.

### USES OF OXYGEN

1. For making oxides e.g. carbon dioxide.
2. For combustion with fuels that provide energy for heating and driving vehicles.
3. For welding. Oxygen can provide very high temperature that can melt metals.
4. For tissue respiration in living things.
5. Used in iron blast furnaces.

2. Particulates affect humans in that they get deposited into vital body organs and block their function.
3. Sulphur dioxide causes irritation of the respiratory tract leading to high mucous secretion.
4. Carbon monoxide is toxic if it is taken by haemoglobin instead of $O_2$.

# CHAPTER 6

## SOIL

- Soil is made up of soil particles, air, water, micro- and macro- organisms and humus.
- **Humus** comes from dead remains of plant and animal matter and it is called ORGANIC MATTER.
- Soil particles come from the process of ***weathering***.

## CHEMICAL PROPERTIES OF SOIL

- Properties include soil pH, Cat ion Exchange Capacity (CEC), salinity and organic matter content.

1. SOIL pH
   - This measures the acidity or alkalinity of the soil.
   - Knowledge of soil **pH** determines the type of crops to be grown in a particular area.
   - pH means power of hydrogen ions in a solution.

### EFFECTS OF SOIL pH

a. It affects the availability of plant nutrients. In acidic soils, mineral salts become soluble and are easily washed away.
b. Soil pH affects the availability of micro-organisms in the soil.



### FACTORS THAT AFFECT SOIL pH

- Application of acidifying fertilizers: phosphate and sulphate of ammonium add acid into the soil.
- Leaching: heavy rains cause $Ca^{2+}$, $Na^{+}$, $Mg^{2+}$ and $K^{+}$ to go down into the soil leaving the concentration of $H^{+}$ to increase thereby making soil acidic.
- Parent material: sulphur parent material produces acidic soil. The acidity is caused by the $H_2SO_4$ [sulphuric acid] which is formed from sulphur. Limestone [$CaCOH_2$] releases alkaline soil.
- Drainage: sandy soils tend to have lower PH [acidic] while clay soils have higher PH [alkaline].

### IMPORTANCE OF CONTROLLING SOIL ACIDITY

- This is important because it creates a good environment for plant or crop production and availability of micro-organisms. Presence of micro-organisms improves ***aeration*** of the soil and they ***enrich*** the soil with their waste products.

2. CATION EXCHANGE CAPACITY (CEC)
- This is the ability of the soil to exchange ions at a given PH per unit weight. Sometimes ions are applied into the soil to take place of those present in it. A good example is the ***liming process***. When liming, calcium oxide is applied to the acidic soil. Calcium ions replace the hydrogen ions in the acidic soil.

3. SOIL SALINITY
- This means the salt content of the soil.
- Salinity means the salt content of the soil. The accumulation of salts in the soil is called SOIL SALINATION. Saline soil usually contains a lot of sulphates [$NO_4^{2-}$], nitrates [$NO_3^{-}$], carbonates [$CO_3^{2-}$] and chlorine [$Cl^{-}$].

### IDENTIFICATION OF SALINE SOILS

- Release of salty water from a borehole.
- Accumulation of white substances on the surface of the soil.

- Goats and cattle licking the soil.

**EFFECTS OF SALINE SOILS**

- It raises the PH of the soil [alkaline].
- Seed germination and plant growth usually fail because water is not available to plants and seeds due to high concentration of salts in the soils.
- It reduces the presence of micro-organisms in the soil.

4. ORGANIC MATTER

➢ This is the decaying plant and animal remains in the soil.

**IMPORTANCE OF ORGANIC MATTER**

- It binds soil particles together thereby improving drainage, aeration and reducing soil erosion.
- It improves soil structure and water holding capacity of the soil.
- Promotes the activity of the micro-organisms.

**SOIL POLLUTION**

➢ **Soil pollution** is the introduction of substances that are harmful to the soil. It may be due to the human and industrial activities. Substances that encourage pollution are called *POLLUTANTS*.

**SOURCES OF SOIL POLLUTANTS AND DEGRADATION**

The following are human activities which encourage soil pollution:

1. Poor farming methods; e.g. making ridges along the slope.
2. Application of inorganic fertilizers and pesticides. Inorganic fertilizers add acidity into the soil and some pesticides are non-biodegradable.
3. Car exhausts and industrial waste such as $CO_2$, $NO_2$ and $SO_2$. When these dissolve in water they form acid rain which increases soil acidity.



4. The reactions are as follows:
   a. $CO_2$ (g) + $H_2O$ (l) → $H_2CO_3$ (aq) Carbonic acid
   b. $4NO_2$ (g) + $2H_2O$ (l) → $4HNO_2$ (aq) Nitric acid
   c. $SO_2$ (g) + $H_2O$ (l) → $H_2SO_4$ (Sulphuric acid)

5. Waste plastic and industrial effluent deposits. Waste plastic is non-biodegradable and remains in the soil for a long time while effluent kills marine organisms.
6. Lead compounds: these are released from exhaust pipes and when dumped into the soil, they kill soil organisms.

**PREVENTION OF SOIL POLLUTION AND DEGRADATION**

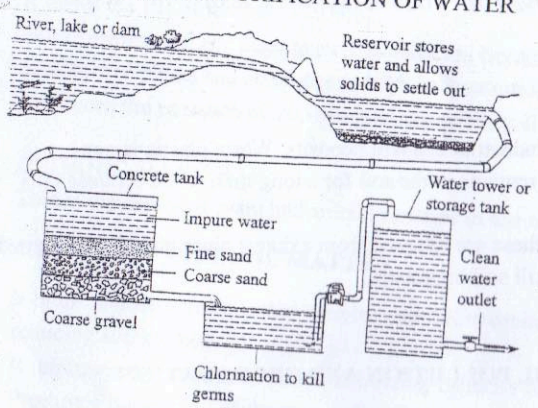
Several methods are used:

- Using electricity and solar power maintains soil cover.
- Using good farming methods or practices.
- Afforestation and Re-afforestation.
- Recycling industrial and domestic wastes.
- Processing the wastes in fuels.
- Application of organic fertilizers like manure and animal dung.

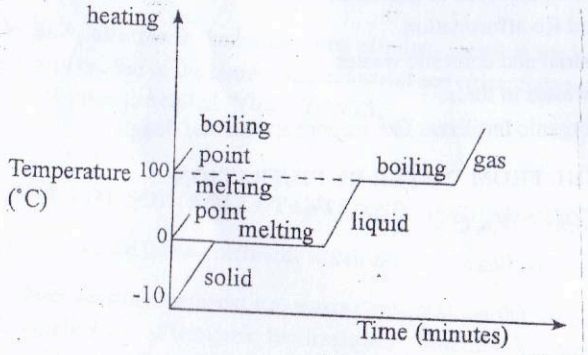
**SEPARATION OF SOIL FROM WATER BY FILTRATION**

SEE PAGE 23

## MAIN STAGES IN THE PURIFICATION OF WATER




## EFFECT OF HEAT ON TEMPERATURE





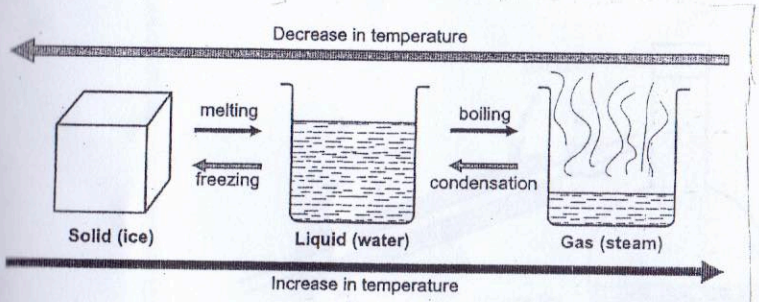




*Fig. 3.17 Effect of heat on ice*